

→ find out the list of companies which are conducting the placement ^{drive} using DSA course

→ find out the syllabus of gate for DSA.

→ find out set of i/o for the following

① Electricity bill

② Bus ticket

③ Message to a friend

→ today when you started from ^{hostel} home to college what are the various data structures you identified in your surroundings.

gate syllabus

1. Programming & Data Structures

- C programming basics
- Recursion
- Arrays
- Stacks
- Queue
- Linked Lists
- Trees
- Binary Search Trees
- Binary Heaps
- Graphs

2. Algorithms

- Searching
 - Linear search
 - Binary search
- Sorting - Bubble sort, selection sort, Insertion sort, Merge sort, Quick sort, Heap sort
- Hashing
- Time and space complexity
 - Big- $O(n)$
 - Omega(Ω)
 - Theta(Θ)
- Algorithm Design Techniques
 - Divide and conquer
 - Greedy Algorithms
 - Dynamic programming
- Graph Algorithms
 - BFS
 - DFS
 - Minimum spanning tree
 - Shortest path

Tier 1 (very high DSA)

- Google
- Microsoft
- Amazon
- Meta
- Apple
- NVIDIA
- Uber
- Atlassian
- Adobe
- Salesforce

Tier 2 (Strong DSA)

- Flipkart
- Walmart
- Intuit
- Oracle
- Paypal
- Visa
- American Express
- Goldman Sachs
- Morgan Stanley
- Deloitte

Indian product companies

- iPhonepe
- Razorpay
- Meesho
- Groww
- CRED
- Freshworks
- Zoho
- Juspay
- Swiggy
- Zomato

Service Based companies

(Basic to moderate DSA)

- TCS
- Infosys
- Wipro
- Cognizant
- Accenture
- Capgemini
- HCLTech
- Tech Mahindra
- IBM

③ Electricity bill:

Input:

- Consumer Number
- Customer Name
- Previous Meter Reading
- Current Meter Reading
- Rate per unit
- Fixed charge

Output:

- Units Consumed
- Electricity Bill Amount
- Due date
- Payment status

2) Bus ticket

Input:

- Passenger Name
- Source
- Destination
- Date
- Bus Type
- Number of Tickets.

Output:

- Ticket Number
- Passenger Name
- Seat Number
- Fare
- Bus Number
- Journey Date

3) Sending a message to friend.

Input:

- Sender Name
- Receiver Name
- Message
- Date
- Time

Output:

- Receiver
- Message
- Date & Time
- Status

④ Mess token

plates

classroom benches

Books

Subjects

attendance register

~~array~~ object

array

array

array

array

array